

Practical-9

Aim

To analyze large-scale social media hashtag data using Apache Spark in order to identify trending hashtags, understand user interests, compare trends across multiple days and regions, categorize hashtags, and generate useful insights for digital marketing and business decision-making.

Problem Definition:

You are a Big Data Analyst at TrendPulse Social Media Analytics, a company that monitors social media activity across multiple platforms such as Twitter, Instagram, Facebook and LinkedIn. Every day, millions of users post content containing hashtags related to current events, technology, sports, entertainment, politics, and business. The marketing team wants to identify: most trending hashtags, popular discussion topics, user interests, and emerging market trends. Traditional analytics tools cannot efficiently process the massive volume of social media data generated every day.

Objectives

1. To create a Spark Session for processing social media data.
2. To load and validate a social media hashtag dataset using Spark DataFrames.
3. To group identical hashtags and calculate their frequency.
4. To identify the top 3 trending hashtags based on their occurrence.
5. To compare hashtag trends across multiple days.
6. To analyze hashtag popularity across different regions.
7. To classify hashtags into categories such as Technology, Sports, and Entertainment.
8. To generate visualizations for hashtag trends and category popularity.
9. To interpret user interests and emerging trends.
10. To provide business and digital marketing recommendations based on the analysis.

Objective/Output:

Tasks: Create Spark Session.

TASK 1: SPARK SESSION

Spark Session Created Successfully
Spark Version: 4.2.0

Load social media hashtag dataset.

TASK 2: LOAD DATASET

Dataset Loaded Successfully

Display hashtag records.

TASK 3: DISPLAY HASHTAG RECORDS

date	platform	region	hashtag
2026-10-01	Twitter	India	#AI
2026-10-01	Instagram	India	#AI
2026-10-01	Facebook	USA	#Sports
2026-10-01	LinkedIn	India	#Technology
2026-10-01	Twitter	USA	#Sports
2026-10-01	Instagram	UK	#Entertainment
2026-10-01	Facebook	India	#AI
2026-10-01	Twitter	India	#Technology
2026-10-01	LinkedIn	USA	#Business
2026-10-01	Instagram	India	#Sports
2026-10-02	Twitter	India	#AI
2026-10-02	Instagram	India	#AI
2026-10-02	Facebook	USA	#AI
2026-10-02	LinkedIn	India	#Technology
2026-10-02	Twitter	USA	#AI
2026-10-02	Instagram	UK	#Entertainment
2026-10-02	Facebook	India	#AI
2026-10-02	Twitter	India	#AI
2026-10-02	LinkedIn	USA	#Business
2026-10-02	Instagram	India	#Technology

only showing top 20 rows

Validate dataset structure.

TASK 4: VALIDATE DATASET STRUCTURE

Schema:

root

```
-- date: date (nullable = true)
-- platform: string (nullable = true)
-- region: string (nullable = true)
-- hashtag: string (nullable = true)
```

Columns:

['date', 'platform', 'region', 'hashtag']

Total Records:

40

Missing Values:

```
+-----+-----+-----+-----+
|date|platform|region|hashtag|
+-----+-----+-----+-----+
|  0 |        0 |      0 |      0 |
+-----+-----+-----+-----+
```

Group identical hashtags.

TASK 5: GROUP IDENTICAL HASHTAGS

```
+-----+-----+
|      hashtag|count|
+-----+-----+
|          #AI|    20|
|  #Technology|     7|
|        #Sports|     6|
| #Entertainment|     4|
|        #Business|     3|
+-----+-----+
```

Count hashtag occurrences.

TASK 6: COUNT HASHTAG OCCURRENCES

```
+-----+-----+
|      hashtag|occurrences|
+-----+-----+
|          #AI|          20|
|  #Technology|           7|
|        #Sports|           6|
| #Entertainment|           4|
|        #Business|           3|
+-----+-----+
```

Sort hashtags based on popularity.

TASK 7: SORT HASHTAGS BY POPULARITY

hashtag	occurrences
#AI	20
#Technology	7
#Sports	6
#Entertainment	4
#Business	3

Identify top trending hashtags.

TASK 8: TOP 5 TRENDING HASHTAGS

hashtag	occurrences
#AI	20
#Technology	7
#Sports	6
#Entertainment	4
#Business	3

Generate trend analysis report.

Unique Hashtags: 5
Total Records: 40

Interpret user interests.

Interpretation of User Interests

The hashtag analysis helps identify the topics that users are most interested in based on hashtag frequency. Highly frequent hashtags indicate strong user engagement and popular areas of discussion. Technology-related hashtags such as **#AI** and **#Technology** indicate interest in emerging technologies and innovation. Sports-related hashtags indicate interest in sports and related events, while entertainment hashtags represent interest in entertainment content.

The comparison of hashtags across different days helps identify whether a topic is becoming more or less popular. Regional analysis can further show differences in user interests between regions. Therefore, hashtag frequency, daily trends, and regional popularity can be used to understand changing user preferences and emerging topics.

Suggest business recommendations.

1. **Target trending topics:** Businesses should use popular hashtags in relevant marketing campaigns to increase visibility and engagement.
2. **Create personalized campaigns:** Regional hashtag analysis can help businesses create location-specific marketing strategies based on local user interests.
3. **Focus on emerging trends:** Businesses should monitor hashtags whose popularity is increasing rapidly and develop products, services, or content around emerging interests.
4. **Improve social media content:** Companies can use popular topics to create relevant posts, advertisements, videos, and promotional content.
5. **Use platform-specific strategies:** Platform-wise hashtag analysis can help businesses understand which topics are popular on different social media platforms and optimize their campaigns accordingly.
6. **Real-time trend monitoring:** Businesses should continuously monitor social media trends so they can respond quickly to changing customer interests, market opportunities, and potential negative feedback.
7. **Support data-driven decisions:** Hashtag analytics can help marketing teams make decisions based on actual social media activity rather than assumptions.

Key Questions / Analysis

Q1. Why is Apache Spark suitable for social media analytics?

Apache Spark is suitable because social media platforms generate huge volumes of data every day. Spark can process large datasets quickly using distributed and in-memory processing. It also supports DataFrames, SQL, streaming, and machine learning, making it useful for analyzing hashtags, posts, users, and trends efficiently.

Q2. What is the significance of hashtag frequency?

Hashtag frequency indicates how often a particular topic or keyword is being discussed on social media. A high-frequency hashtag may represent a popular or trending topic. By calculating hashtag frequency, businesses can identify popular discussions, user interests, and emerging topics.

Example: If #AI occurs much more frequently than other hashtags, it indicates strong interest in Artificial Intelligence.

Q3. How can trend analysis support digital marketing campaigns?

Trend analysis helps marketers understand what people are currently interested in. Businesses can use trending hashtags and topics to create relevant advertisements, social media posts, promotional campaigns, and content.

For example, if a particular technology-related hashtag suddenly becomes popular, a technology company can create marketing content around that topic to attract more users.

Q4. Why do businesses monitor social media trends?

Businesses monitor social media trends to:

- Understand customer interests and preferences
- Identify emerging market opportunities
- Track competitors and industry discussions
- Improve marketing strategies
- Detect changes in customer sentiment
- Identify popular products or topics
- Respond quickly to changing customer demands

Therefore, social media trends provide valuable information for data-driven business decisions.

Q5. How can real-time analytics improve decision making?

Real-time analytics allows businesses to analyze social media activity as it happens rather than waiting for a daily or weekly report.

For example, if a product suddenly becomes popular or receives negative feedback, the company can immediately respond by changing its marketing strategy, addressing customer complaints, or increasing product availability.

Thus, real-time analytics helps businesses make faster, more accurate, and timely decisions.

Supplementary Problems:

Identify top 3 trending hashtags.

```
=====
SUPPLEMENTARY TASK 1: TOP 3 TRENDING HASHTAGS
=====
+-----+-----+
| hashtag | occurrences |
+-----+-----+
| #AI      | 20          |
| #Technology | 7          |
| #Sports  | 6           |
+-----+-----+
```

Compare trends across multiple days.

```
=====
SUPPLEMENTARY TASK 2: MULTI-DAY TREND ANALYSIS
=====
+-----+-----+-----+
| date   | hashtag | occurrences |
+-----+-----+-----+
| 2026-10-01 | #AI      | 3           |
| 2026-10-01 | #Sports  | 3           |
| 2026-10-01 | #Technology | 2          |
| 2026-10-01 | #Business | 1           |
| 2026-10-01 | #Entertainment | 1          |
| 2026-10-02 | #AI      | 6           |
| 2026-10-02 | #Technology | 2          |
| 2026-10-02 | #Business | 1           |
| 2026-10-02 | #Entertainment | 1          |
| 2026-10-03 | #AI      | 4           |
| 2026-10-03 | #Sports  | 3           |
| 2026-10-03 | #Technology | 2          |
| 2026-10-03 | #Entertainment | 1          |
| 2026-10-04 | #AI      | 7           |
| 2026-10-04 | #Entertainment | 1          |
| 2026-10-04 | #Technology | 1           |
| 2026-10-04 | #Business | 1           |
+-----+-----+-----+
```

Analyze hashtag popularity by region.

```
=====
SUPPLEMENTARY TASK 3: REGIONAL ANALYSIS
=====
```

region	hashtag	occurrences
India	#AI	15
India	#Technology	7
India	#Sports	2
UK	#Entertainment	4
USA	#AI	5
USA	#Sports	4
USA	#Business	3

Create hashtag categories (Technology, Sports, Entertainment).

```
=====
SUPPLEMENTARY TASK 4: HASHTAG CATEGORIES
=====
```

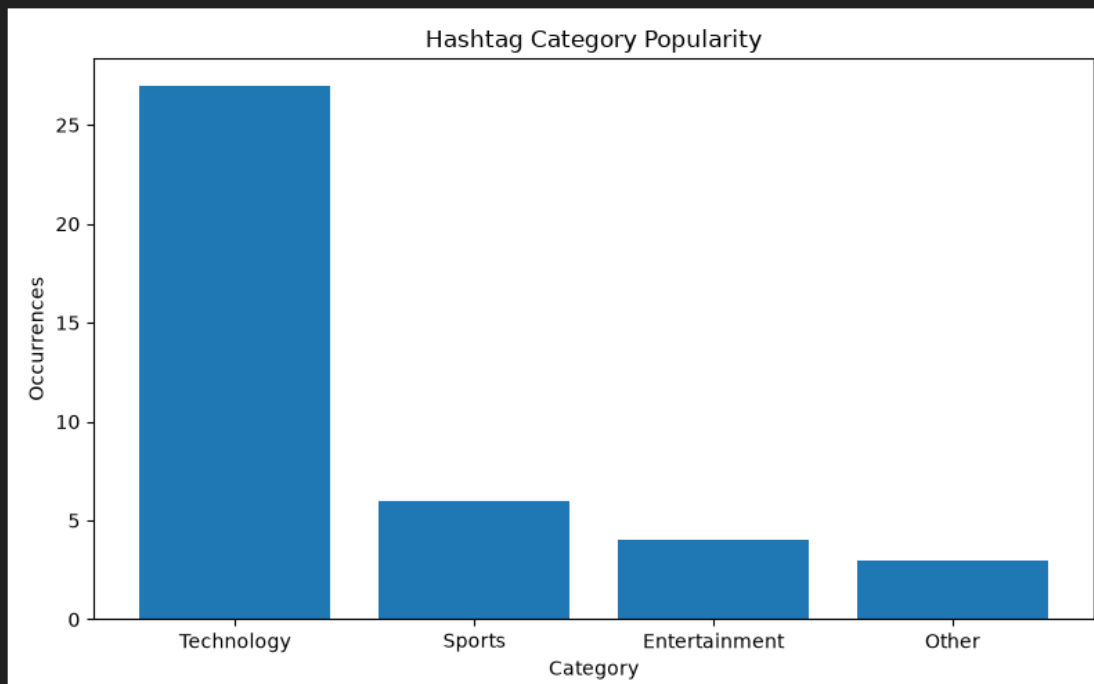
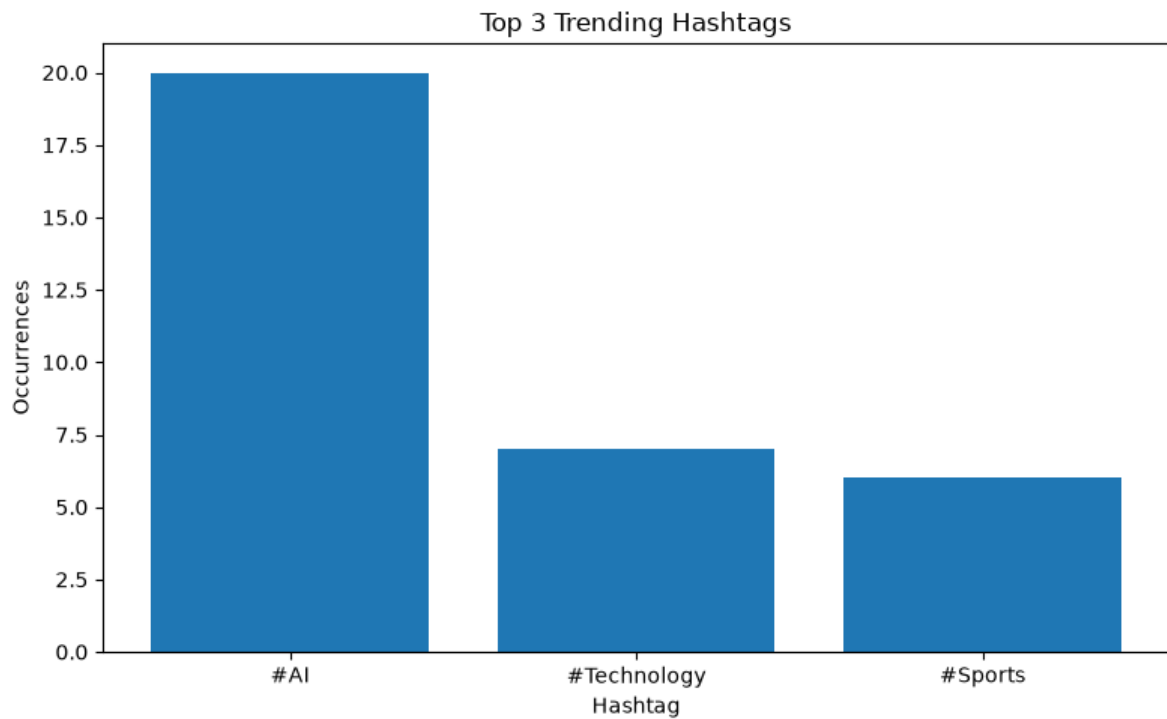
date	platform	region	hashtag	category
2026-10-01	Twitter	India	#AI	Technology
2026-10-01	Instagram	India	#AI	Technology
2026-10-01	Facebook	USA	#Sports	Sports
2026-10-01	LinkedIn	India	#Technology	Technology
2026-10-01	Twitter	USA	#Sports	Sports
2026-10-01	Instagram	UK	#Entertainment	Entertainment
2026-10-01	Facebook	India	#AI	Technology
2026-10-01	Twitter	India	#Technology	Technology
2026-10-01	LinkedIn	USA	#Business	Other
2026-10-01	Instagram	India	#Sports	Sports
2026-10-02	Twitter	India	#AI	Technology
2026-10-02	Instagram	India	#AI	Technology
2026-10-02	Facebook	USA	#AI	Technology
2026-10-02	LinkedIn	India	#Technology	Technology
2026-10-02	Twitter	USA	#AI	Technology
2026-10-02	Instagram	UK	#Entertainment	Entertainment
2026-10-02	Facebook	India	#AI	Technology
2026-10-02	Twitter	India	#AI	Technology
2026-10-02	LinkedIn	USA	#Business	Other
2026-10-02	Instagram	India	#Technology	Technology

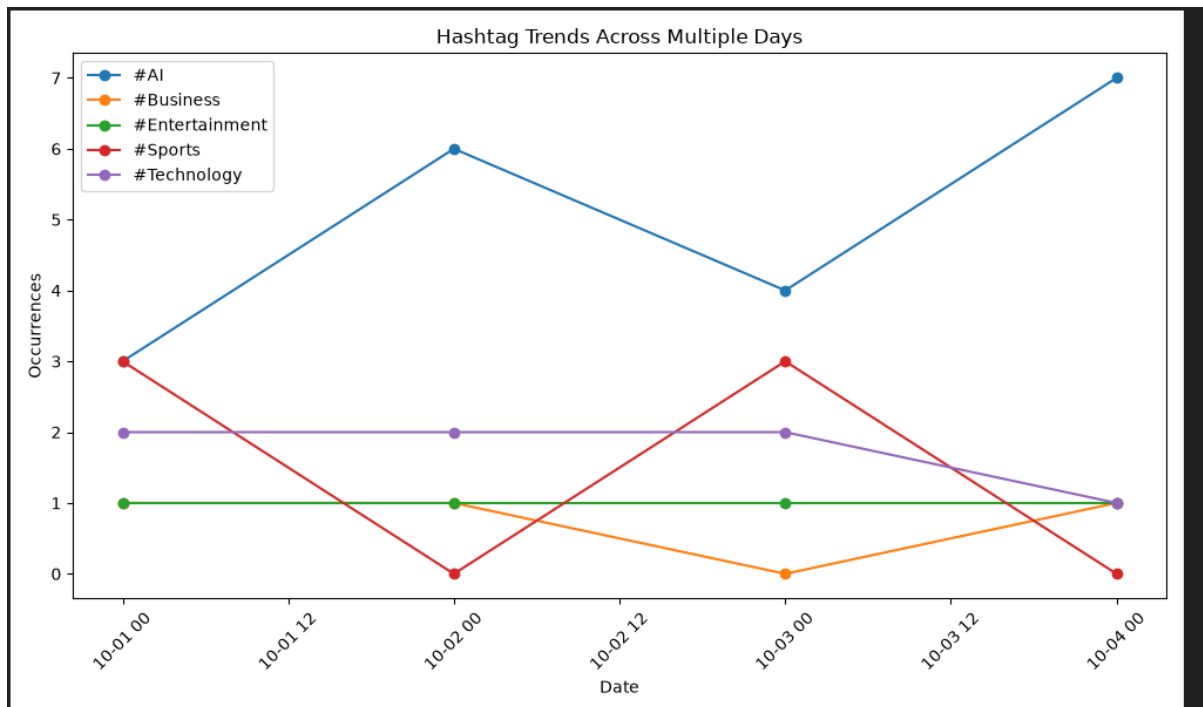
only showing top 20 rows

Category Popularity:

category	occurrences
Technology	27
Sports	6
Entertainment	4
Other	3

Generate trend visualization using charts.





Tools and Technologies Used:

- **Apache Spark** – Large-scale data processing and analysis.
- **PySpark** – Python API for Apache Spark.
- **Python** – Programming language used to implement the analysis.
- **Spark DataFrame** – Used for loading, grouping, filtering, and analyzing the dataset.
- **Pandas** – Used to convert Spark results for visualization.
- **Matplotlib** – Used to generate trend and popularity charts.
- **CSV Dataset** – Used as the social media hashtag data source.
- **Ubuntu/WSL** – Execution environment for the practical.
- **VS Code / Nano** – Used for creating and editing Python and CSV files.

Conclusion:

In this practical, Apache Spark and PySpark were used to analyze social media hashtag data efficiently. The dataset was loaded into a Spark DataFrame, validated, grouped, and analyzed to determine hashtag frequencies and identify the top trending hashtags. Hashtag trends were also compared across multiple

days and regions to understand changes in user interests. Hashtags were categorized into Technology, Sports, and Entertainment categories, and visualizations were generated to represent the results clearly.

The analysis demonstrates how Big Data technologies such as Apache Spark can help organizations process large volumes of social media data and extract meaningful insights. These insights can support digital marketing campaigns, customer-interest analysis, trend detection, and faster data-driven business decision-making.